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ECE 468 Project 1 Report

2 Stage OTA Design

1. Specifications

The given design specifications and requirements for this project were as follows:

$$A_{CL} = 1$$

$$A_{OL} = > 45\text{dB}$$

$$\text{Phase Margin} = > 65^\circ$$

$$\text{Unity-gain Frequency} = > 5\text{MHz}$$

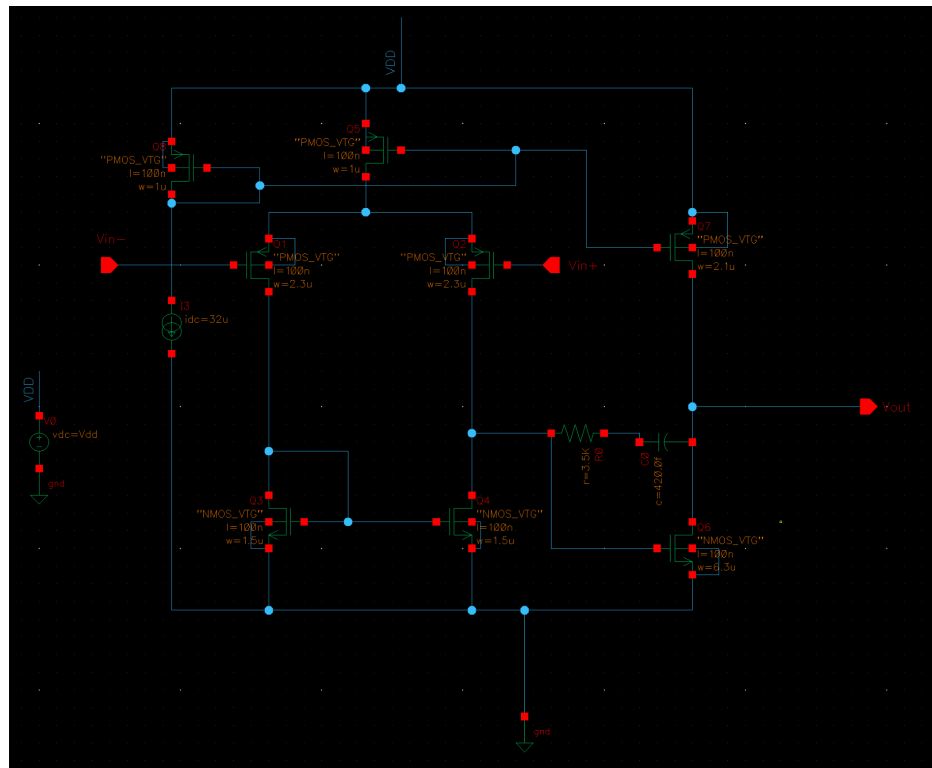
For testing and simulations, the load capacitance was set to 2pF.

$$V_{dd} = 1.1\text{ V}$$

$$\text{Maximum current consumption} = 1\text{mA}$$

2. Topology

Following the topology suggested in the manual, and keeping the transistor lengths to twice the minimum the technology used in this project i.e 50nm, the final design was as follows;



With all the lengths set to 100nm, the other design parameters are shown below:

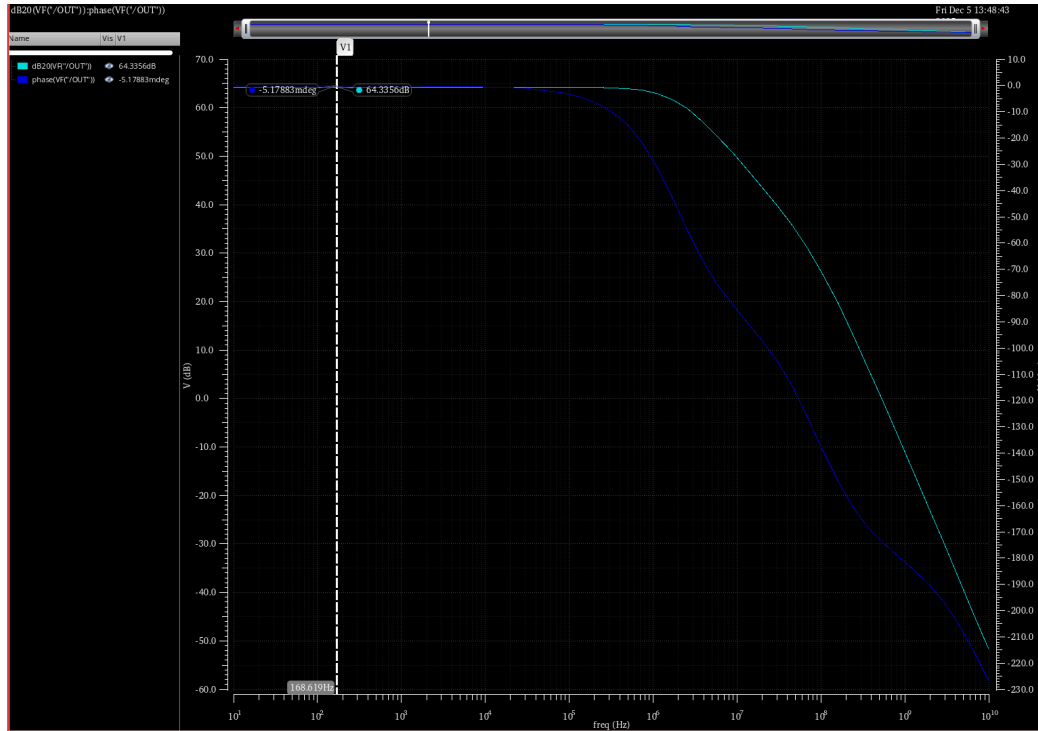
Parameters	Value
I_B	32 μ A
Q5,8 (W)	1 μ m
Q1,2(W)	2.3 μ m
Q3,4 (W)	1.5 μ m
Q7 (W)	2.1 μ m
Q8 (W)	6.3 μ m
C_C	420fF
R_C	3.5k Ω
V_{dd}	1.1 V
V_{CM}	500mV

3. Simulations and Results

a. AC, DC & Transient Analysis

The first simulation run after the design process was AC analysis to determine the open loop gain. The initial stages of this simulation involved setting the length to 50nm and this limited the gain to < 30dB. To compensate for this, an attempt to increase the transconductance was tried by increasing the widths in the first stage but this also required the increase in the biasing current so that all the transistors could operate in saturation hence diminishing its effects. Finally considering low-noise design considerations and overall power consumption, it was decided that setting the lengths to 100nm was the most effective way to increase gain while keeping the biasing current as low as possible. The achieved gain was above 45dB meeting the design specification requirements.

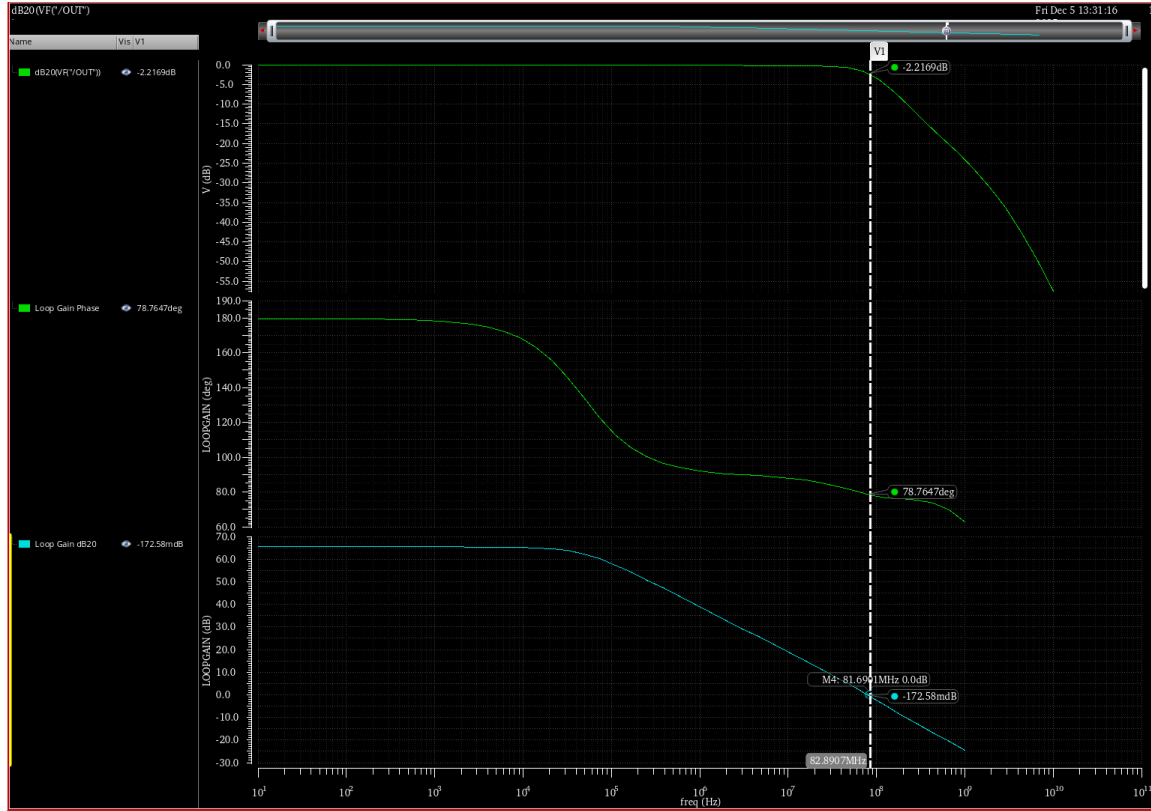
The observed open loop gain curve is as shown:



Alongside AC analysis, DC analysis was run as well to monitor the operating points/regions of the transistors. This analysis showed that all the transistors were operating in saturation and that they were properly biased.

The next analysis was the closed loop frequency response. Before compensation, this simulation showed a spike revealing instability. Following the Miller Compensation Recipe, the first values of C_C and R_C were set to 1.8pF and 5.5k Ω respectively achieving a phase margin of 113°. However during the transient analysis, the simulation results showed that the OTA was too slow hence the need for a lower compensation. The compensation capacitance was reduced to 420fF to increase the slew rate and the resistance to 3.5k Ω to keep the phase margin above 65°. This reduced the phase margin to $\sim 72^\circ$ which was still stable.

Below are the stability simulation results. It includes the output gain, the phase margin and the closed-loop gain.



The final simulation to determine the design specifications was the transient analysis which was used to calculate the slew rate of the OTA and the result is shown below;

Slew Rate	expr	slewRate(VT("/OUT")) 0.5 nil 0.805 nil 10 9...	25.17M	☒
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The table below shows the design goals vs the achieved results.

Parameter	Goal	Obtained Results
A_{OL}	> 45dB	64.33dB
Phase Margin	> 65°	79°
Unity-gain Frequency	5MHz	81.69MHz
Total Current Consumption	< 1mA	129μA
Slew Rate	Not Specified	25.17V/μS

b. Noise Analysis

The simulated total integrated noise in 5MHz bandwidth was 6.544nV^2 and the spot noise at 5kHz was $74.07\text{fV}^2/\text{Hz}$ with the most contribution being from flicker noise from the current mirror Q3 and Q4 as shown:

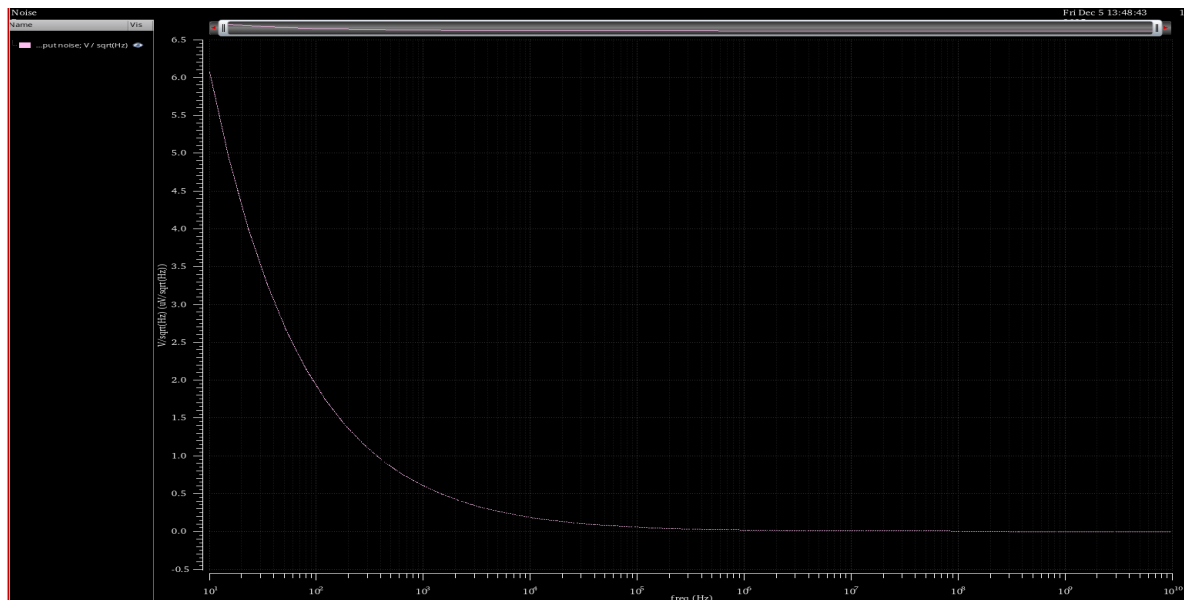
Device	Param	Noise Contribution	% Of Total
/I2/Q4	fn	3.58024e-14	47.17
/I2/Q3	fn	3.33248e-14	44.91
/I2/Q2	fn	2.81597e-15	3.80
/I2/Q1	fn	2.80701e-15	3.78
/I2/Q4	id	6.85643e-17	0.09

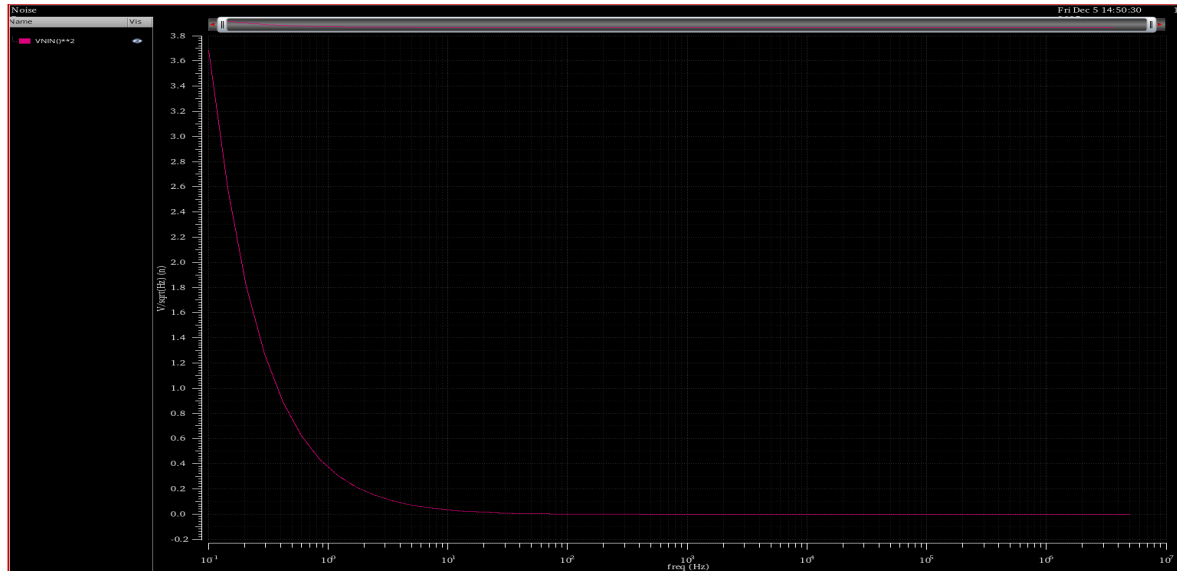
Spot Noise Summary (in V^2/Hz) at 5K Hz Sorted By Noise Contributors
Total Summarized Noise = 7.42002e-14
Total Input Referred Noise = 7.40682e-14
The above noise summary info is for noise data

Device	Param	Noise Contribution	% Of Total
/I2/Q4	fn	2.53482e-09	38.68
/I2/Q3	fn	2.41333e-09	36.83
/I2/Q4	id	3.42682e-10	5.23
/I2/Q3	id	3.26851e-10	4.99
/I2/Q2	id	2.63125e-10	4.02
/I2/Q1	id	2.62612e-10	4.01
/I2/Q2	fn	2.03928e-10	3.11

Integrated Noise Summary (in V^2) Sorted By Noise Contributors
Total Summarized Noise = 6.55299e-09
Total Input Referred Noise = 6.54443e-09
The above noise summary info is for noise data

The power spectral densities of the output noise and the total input-referred noise is shown below respectively.





The flicker noise from these two transistors can be remedied in multiple ways, in this design, increasing the widths of the first stage transistors to increase the drain current noise but at the expense of increasing the current consumption and losing some gain. Increasing the compensation capacitor will also work but this might make the OTA slower.

Conclusion

The initial approach to the design was aimed at keeping the power consumption as low as possible by keeping the transistor sizes smaller. Because of this, the selected final values were based on how small the transistors and the biasing current could get while still operating as expected and meeting the design specifications. While the transistor sizes could get smaller and still meet the specifications, the biasing current used here can not be lowered any further unless the transistor lengths were increased.